

proach, action interface design, context management strategy, and evaluation methodology.

3.1 Architecture and Orchestration

Despite surface-level diversity, autonomous SE agents converge on a small number of architectural patterns. As illustrated in the *Orchestration*, *Action Interface*, and *Context Management* branches of Figure 2, these patterns determine how agents perceive their environment, what actions they can take, and how they manage repository-scale context.

The observe–think–act loop. The dominant paradigm is a ReAct-style [22] loop in which the agent alternates between (i) observing the environment (file contents, test output, error traces), (ii) reasoning about what to do next (chain-of-thought or structured plan), and (iii) executing an action via a tool call. SWE-agent [20] and OpenHands [15] both instantiate this loop, differing mainly in the *action space* exposed to the model (Figure 3).

Agent–Computer Interface (ACI). Yang et al. [20] introduce the ACI concept, arguing that LLMs are a “new category of end users” that need purpose-built interfaces rather than raw shell access. SWE-agent’s ACI includes a windowed file viewer (100 lines), a linter-guarded `edit` command, summarised search results, and compressed observation history. Ablations show that each ACI component contributes measurably: removing the linter drops performance by 3 percentage points; switching to iterative search costs 6 pp; and showing full file content instead of a 100-line window costs 5.3 pp on SWE-bench Lite.

Code-as-action. OpenHands adopts the CodeAct paradigm [14], consolidating all agent actions into executable Python and bash code rather than a fixed tool API. Each turn may run arbitrary IPython cells, bash commands, or browser interactions inside a Docker-sandboxed environment. This design maximises flexibility at the cost of a larger action space that the model must navigate.

Context management. Repository-level tasks easily exceed LLM context windows. Agents employ several mitigation strategies: SWE-agent collapses older observations beyond the last 5 turns; AutoCodeRover [23] parses the codebase into an AST and exposes structure-aware search APIs

(`search_class`, `search_method_in_class`) that return only signatures, not full definitions; Agentless [18] constructs a hierarchical repository skeleton and uses embedding-based retrieval to select candidate files before prompting the LLM.

3.2 Representative Systems

We survey five prominent systems that span the design space. Table 1 provides a comparative overview.

SWE-agent [20]. Developed at Princeton, SWE-agent wraps a GPT-4 Turbo backbone in a custom ACI with specialised file navigation, editing, and search commands. It achieved 12.47% on the full SWE-bench test set (2,294 tasks) and 18.00% on SWE-bench Lite when first published. The agent follows a typical trajectory: reproduce the issue, localise the faulty code via `find_file/search_dir`, then enter an edit–execute loop. The median successful run completes in 12 steps at \$1.21 API cost.

Agentless [18]. From UIUC, Agentless challenges the need for agentic autonomy. It replaces the interactive loop with a fixed three-phase pipeline: *localise* → *repair* → *validate*. Localisation uses a hierarchical narrowing strategy (files → classes/functions → edit locations) combining LLM prompting with embedding retrieval. Repair generates up to 40 candidate patches in a search/replace diff format. Validation selects the best patch via regression testing, LLM-generated reproduction tests, and majority voting over AST-normalised candidates. Agentless resolved 32.00% of SWE-bench Lite at an average cost of only \$0.70 per problem, making it the top open-source system at the time and establishing a strong non-agentic baseline. OpenAI later adopted Agentless as its default scaffold for showcasing GPT-4o and o1 on SWE-bench Verified.

AutoCodeRover [23]. From NUS, AutoCodeRover takes a “software-engineering-oriented” approach, operating on program representations rather than raw files. It parses the codebase into an AST and provides the LLM with structure-aware search APIs. A stratified context retrieval process iteratively expands the agent’s understanding, starting from issue keywords and refining through multiple search rounds until sufficient context is gathered. When a test suite is available, Spectrum-Based Fault Localisation (SBFL)

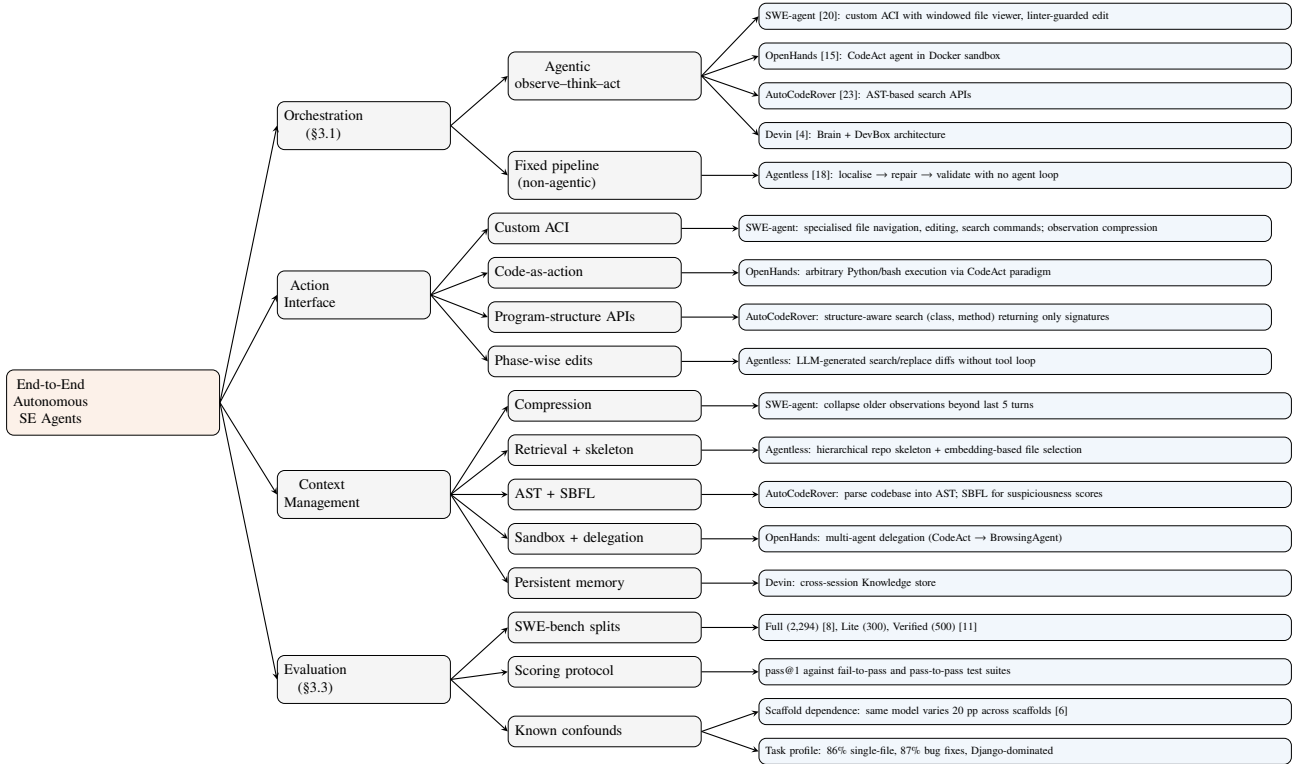


Figure 2 Taxonomy of end-to-end autonomous SE agents. Systems are classified along orchestration approach (agentic loop vs. fixed pipeline), action interface design (custom ACI, code-as-action, structure-aware APIs), context management strategy (compression, retrieval, AST, delegation, memory), and evaluation methodology (benchmark splits, scoring, known confounds). Representative systems are shown as leaf exemplars.

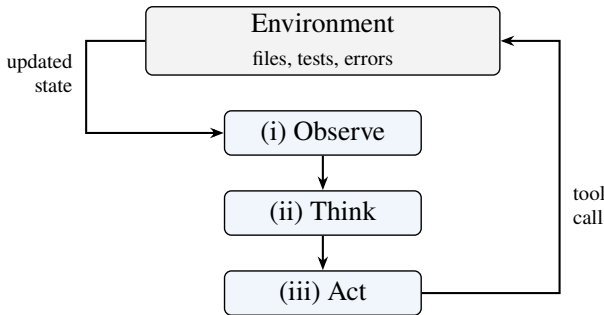


Figure 3 ReAct-style observe-think-act loop. The agent reads environment state, reasons about the next step, executes a tool call, and repeats.

augments the search with suspiciousness scores. AutoCodeRover achieved 19% on SWE-bench Lite in a single run and 26% with three retries, at a cost of \$0.43–\$1.30 per task.

OpenHands [15]. Formerly OpenDevin, OpenHands is an open-source platform (ICLR 2025, 32K GitHub stars, 188 contributors) built around the CodeAct agent. Each session runs inside a Docker container with a bash shell, IPython server, and Chromium browser. The platform supports

multi-agent delegation—a generalist CodeAct agent can offload web browsing to a specialised BrowsingAgent. On SWE-bench Lite, CodeAct v1.8 with Claude 3.5 Sonnet achieved 26.0%. On SWE-bench Verified, a later submission reaches 60.6% with a single rollout and 66.4% when five trajectories are reranked by a dedicated critic model [12].

Devin (Cognition) [4]. Devin is a proprietary cloud-hosted agent positioned as an “autonomous AI software engineer.” Its architecture separates a *Brain* (reasoning layer) from a *DevBox* (execution VM with shell, IDE, and browser). A persistent Knowledge store provides cross-session memory. Cognition reports strong full-benchmark results, but independent leaderboard analysis finds 68.3% on SWE-bench versus 48.2% on SWE-bench Verified (a 29% relative drop), suggesting possible overfitting to easier unfiltered instances [5].

Taken together, these five systems expose a recurring trade-off between agentic flexibility and pipeline efficiency. Agentless shows that a fixed localise–repair–validate pipeline can outperform interactive agents on SWE-bench Lite at lower cost (\$0.70 vs. \$1.21), chal-

Table 1 Comparison of end-to-end autonomous SE agents. Architectural patterns: **OTA** = observe–think–act loop, **ACI** = custom Agent–Computer Interface, **CaA** = code-as-action, **Pipe** = fixed multi-phase pipeline (no agent loop). SWE-bench scores are single-pass (pass@1) unless noted; **Lite** = SWE-bench Lite (300 instances), **Full** = SWE-bench Full (2,294 instances). Cost is average USD per task instance.

System	Origin	Architecture	Context Strategy	Lite	Full	Cost
SWE-agent	Princeton	OTA + ACI	Observation compression	18.0%	12.5%	\$1.21
Agentless	UIUC	Pipe	Hierarchical skeleton + embeddings	32.0%	–	\$0.70
AutoCodeRover	NUS	OTA + AST search	AST-based APIs + SBFL	19% [†]	12.4%	\$0.43
OpenHands	Multi-inst.	OTA + CaA	Docker sandbox + delegation	26.0%	–	–
Devin	Cognition	OTA + Brain/DevBox	Persistent knowledge store	– [‡]	–	propr.

[†]26% with 3 retries (pass@3). [‡]Proprietary; see text for Verified estimates.

lenging the assumption that autonomy is necessary for strong results. Among agentic designs, interface and context strategy matter as much as the backbone LLM: SWE-agent’s ACI ablations, OpenHands’s CodeAct action space, and AutoCodeRover’s AST search APIs each demonstrate that how an agent reads and edits its code can shift resolve rates by several percentage points at comparable model capacity.

3.3 Full-Pipeline Evaluation

The *Evaluation* branch of Figure 2 summarizes the primary benchmark splits, scoring protocols, and known confounds that shape current performance reports.

SWE-bench: dataset construction. The primary evaluation instrument for end-to-end agents is SWE-bench [8]. It was constructed by scraping merged pull requests from 12 popular open-source Python repositories (including Django, scikit-learn, sympy, matplotlib, and Flask). Each PR was filtered to retain only those that modify the project’s test suite, yielding 2,294 task instances in the **Full** split. For each instance the benchmark provides: (i) the repository snapshot *before* the PR was merged, (ii) the issue description (title + body), (iii) a gold patch (the human-written fix), and (iv) a set of *fail-to-pass* tests (which fail before the fix and should pass after) and *pass-to-pass* tests (which must continue to pass). An agent receives only items (i) and (ii); its generated patch is applied and scored against both test sets.

Splits and filtering. Three subsets are commonly used (Table 2):

SWE-bench Lite selects 300 instances for faster iteration but was found to contain quality issues: Xia et al. [18] report that 4.3% of instances contain the exact gold patch in the issue text, 9.7% provide complete solution steps in natural language, and 10% lack

Table 2 SWE-bench dataset splits.

Split	Size	Selection criterion
Full	2,294	All filtered PRs
Lite	300	Subset selected for tractability
Verified	500	Human-validated

sufficient information to be solvable. **SWE-bench Verified** [11] addresses these concerns: three independent human annotators assessed each Full instance for problem clarity, test correctness, and overall solvability, retaining 500 high-quality instances. The Epoch AI analysis [6] estimates a remaining error rate of 5–10%. Verified has become the de facto standard for leaderboard comparisons.

Task complexity profile. Although SWE-bench tests real-world issues, most tasks are relatively small [7, 6]. In the Verified split, human annotators estimated that 39% of tasks are trivial (<15 min fix), 52% are small (15 min–1 h), 8% are substantial (1–4 h), and only 3 tasks were rated >4 h. These estimates correlate with patch size: easy tasks average 5 changed lines of code (LOC) and 1.0 files; medium tasks average 14 LOC and 1.3 files; hard tasks average 56 LOC and 2.0 files. Crucially, 86% of instances are single-file changes, and 87% are classified as bug fixes rather than feature requests or refactorings. The dominance of Django (nearly 50% of instances) and four other repositories (>80% combined) further limits diversity.

Scaffold dependence. An important confound is that *performance reflects scaffold design as much as model capability* [6]. The same model can vary by 20 pp depending on the scaffold: for example, Claude 3.7 Sonnet scores 62.3% with one scaffold and 70.2% with another; GPT-4o scores 23% under SWE-agent but 33.2% under Agentless. This makes

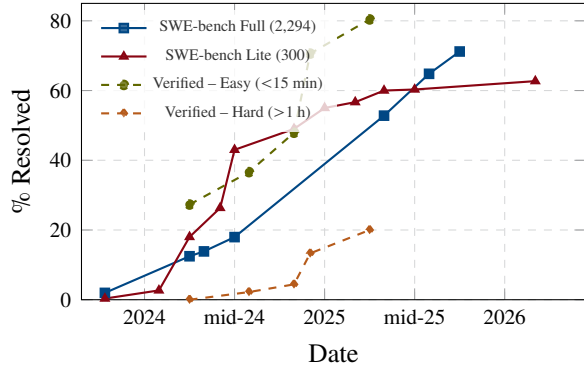


Figure 4 SWE-bench progress over time. Solid lines: best-in-class resolve rate on the Full set (blue, 2,294 tasks) and Lite (red, 300 tasks). Dashed lines: performance on easy (<15 min, green) vs. hard (>1 h, orange) subsets of Verified, illustrating the widening difficulty gap [7]. Data compiled from leaderboard submissions [13, 5, 1].

cross-system comparisons difficult unless the scaffold is controlled, motivating standardised evaluation harnesses such as Live-SWE-agent [19].

Leaderboard trajectory. Figure 4 charts the rapid progress on SWE-bench from late 2023 through mid-2026. In October 2023, the best RAG baselines resolved 1.96% of Full and 0.33% of Lite [8]. SWE-agent’s agentic loop lifted Full to 12.5% and Lite to 18.0% in early 2024. On the Full set, AutoCodeRover reached 18.0% (pass@3) and Devin 13.9% around the same period. The scaffolding race then accelerated both tracks: by late 2024, Agentless 1.5 with Claude 3.5 Sonnet reached 40.7% on Lite; by mid-2025 the best Lite score stood at 60.3%. On the Full 2,294-instance set, fewer teams report results post-2024, but dataku’s independent analysis [5] records Claude Opus 4 at 71.2% and Devin at 68.3%. On Lite, the frontier reached ~63% by early 2026 [13].

The easy–hard gap is striking: the best individual system solves ~80% of easy tasks (<15 min) but only ~20% of hard ones (>1 h) [7]. Combined across all top systems, 95% of easy instances have been resolved by at least one agent, versus only 42% of hard ones—and among hard multi-file issues, only 9 of 25 have ever been solved. This ceiling effect, combined with contamination risk (half the issues predate 2020 and the repositories are widely used as training data [6]), raises concerns about the benchmark’s continued discriminative power. SWE-bench Pro, a harder leaderboard with novel uncontaminated problems, shows Claude Opus 4.5 dropping to ~46% [1].

Beyond SWE-bench. SWE-bench covers only Python bug-fixing in 12 repositories. Newer benchmarks begin to address broader scopes: SWE-bench Multilingual (300 tasks, 9 languages), CodeAgent-Bench (interactive development environments), and real-world deployment metrics. A comprehensive discussion of the evaluation landscape is deferred to §7.2.

4 Planning and Search Strategies for Code Synthesis

Planning is the stage in which an agent turns a vague intent into an operational hypothesis about *what* code should change and *how* the change can be produced. In the vibe coding pipeline, this stage sits between intent capture and code generation, but in practice it is rarely a single up-front step. Agents plan while reading issue reports, searching repositories, choosing files, decomposing edits, sampling candidate patches, and revising their strategy after failed tests. Thus, planning is best understood as a control layer over generation rather than as a separate document written before generation begins.

This section surveys three aspects of that control layer:

1. Plan representations: determine what intermediate state the agent exposes to itself: natural-language rationales, hierarchical task lists, program sketches, or repository-localisation hypotheses;
2. Search strategies: determine how many possible solutions are explored and how candidates are ranked;
3. The plan-generate interface: determines whether planning happens once, repeatedly, or only when execution feedback invalidates the current hypothesis.

4.1 Plan Representations and Decomposition

Plan representations determine how an agent organises reasoning before generating code. The simplest is chain of thought [17]: a linear sequence that commits early to one trajectory. If the first design choice is wrong, subsequent steps elaborate the mistake. Tree of Thoughts [21] and Graph of Thoughts [2] relax this by allowing exploration, backtracking, and merging across alternative paths (Figure 5). For code synthesis, this means agents can consider multiple decomposition

Table 3 Planning and search mechanisms in LLM-based code synthesis.

Mechanism	Planning object	Role in code tasks
Chain / tree / graph of thoughts	Intermediate reasoning steps	Explore alternative designs, algorithms, or debugging hypotheses before emitting code
Hierarchical decomposition	Subtasks, files, functions, edit locations	Reduce repository-scale tasks into bounded local edits
Candidate sampling	Multiple complete or partial programs	Trade inference cost for higher pass@k and diversity
Execution-guided selection	Tests, generated tests, compiler output	Rank or prune candidates using external evidence rather than model confidence alone
ReAct-style re-planning	Updated belief state after tool observations	Interleave search, reading, editing, and feedback in a single loop

strategies—“change the parser”, “change the caller”, or “adjust the compatibility layer”—as explicit search states rather than implicit in a single sample.

Repository-level agents usually add a more concrete representation: *localisation plans*. Agentless decomposes issue resolution into files, classes/functions, and edit locations before generating a patch [18]. AutoCodeRover uses AST-backed APIs such as `search_class` and `search_method_in_class` to plan over program structure rather than raw text [23]. These grounded representations name concrete program elements that can be validated against the repository (Figure 6), turning a broad intent into a bounded search problem.

Plans may also be expressed as *specification sketches*: partial interfaces, pseudocode, assertions, or examples that constrain the generated implementation. Useful plans are grounded in observable artefacts—issue text, existing code, API conventions, and failing tests—rather than invented requirements.

4.2 Search and Exploration Strategies

Once a representation exists, an agent must decide how broadly to search. Single-sample generation is fast, but it treats the model’s first completion as the solution. Search-based systems instead spend additional inference or execution budget to explore alternatives. The simplest form is self-consistency: sample multiple reasoning paths and select the most common or most consistent answer [16]. For programming, the same idea becomes sample-and-rank: generate several candidate programs or patches, then choose using tests, static checks, model judges, or majority voting.

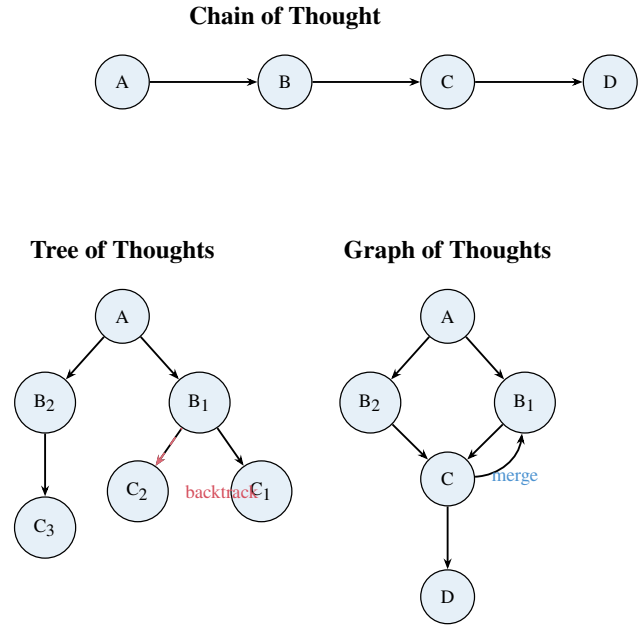


Figure 5 Plan representation structures. Chain of Thought commits to a single linear path. Tree of Thoughts explores alternatives and backtracks. Graph of Thoughts merges branches and revisits earlier states.

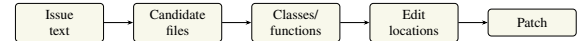


Figure 6 Hierarchical localisation pipeline. Agents narrow scope iteratively from repository-level to line-level edits.

AlphaCode [9] and CodeT [3] exemplify this pattern (Figure 7). AlphaCode generated millions of candidates for competitive programming, filtered and clustered by execution results, then submitted a small diverse set. CodeT generates both programs and tests, selecting code consistent with generated test cases. The key insight is that LLMs often contain the right solution somewhere in the sample distribution even when top-ranked output is wrong. The limitation is self-referential objectives: if generated tests are weak, selection optimises for the model’s own misunderstanding. Execution-guided search is strongest when feedback includes external oracles—existing unit tests, compiler errors, type checkers, or repository-specific regression suites.

Search also appears in repository navigation. Agentless samples multiple locations and candidate patches, then normalises patches by AST and uses validation to pick among them [18]. AutoCodeRover expands context iteratively through program-structure queries, optionally using Spectrum-Based Fault Localisation (SBFL) to prioritise suspicious code when tests are available [23]. In both cases, search is not over complete programs alone; it is also over *where to look* and

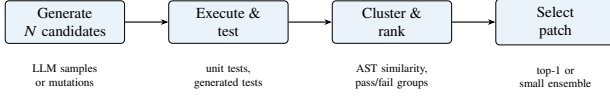


Figure 7 Candidate-selection pipeline. Systems like AlphaCode and CodeT generate many candidates, execute them against tests (real or generated), cluster by behaviour, and select a small high-quality set.

Table 4 Plan–generate coupling patterns.

Pattern	Mechanism	Trade-off
Plan-then-generate	Plan once, then generate code conditioned on it	Easy to inspect; brittle if plan is wrong
Interleaved planning	Alternate plan updates and tool use in observe–think–act loop	Adaptive; plan distributed across turns
Generate-then-refine	Generate, critique via feedback, revise iteratively	Self-correcting; may reopen planning after execution

what evidence to include. This matters because code generation quality is bounded by context quality. A correct patch is unlikely if the agent never places the relevant function, invariant, or failing trace in context. The cost of search is the main trade-off. More samples increase diversity and pass@k, but they also increase latency, token cost, and validation cost. Moreover, search can overfit to narrow tests: a candidate that passes the available suite may still violate unstated requirements. Strong systems therefore combine multiple pruning signals: syntactic validity, static analysis, failing-test repair, pass-to-pass regression preservation, and semantic similarity to the intended behaviour. Search is most useful when the ranking signal is at least as reliable as the generator is diverse.

4.3 Plan–Generate Interplay

Planning and generation can be coupled in three broad ways (Table 4). *Plan-then-generate* is easy to inspect and aligns with human practice, but brittle if the initial plan is wrong [18]. *Interleaved planning* adapts to new information but distributes the plan across many turns, making it harder to audit [22]. *Generate-then-refine* treats the first output as a hypothesis and revises via self-critique or execution feedback [10], connecting planning to the refinement loop in §6.

The central design question is not whether agents should plan, but how plans remain *grounded*. Ground-

ing mechanisms include retrieval over repository structure, AST-level symbol search, execution traces, failing tests, and explicit constraints extracted from the issue report. Effective systems keep plans close to these artefacts and treat generation as one move in a larger search process, reframing code synthesis from “write the answer” to “maintain and test a hypothesis about the required program change”.

5 Multi-Agent Frameworks for Code Generation

An alternative architectural approach to the generation-to-validation span: multiple specialized agents divide labor by role (architect, coder, tester, reviewer), collectively covering planning, generation, and internal verification to improve code quality beyond what a single agent achieves.

5.1 Role Decomposition and Communication

How tasks are split among agents, what roles exist, and how agents exchange information (shared blackboard, message passing, structured dialogue).

5.2 Framework Designs

Survey of concrete frameworks (e.g., ChatDev, MetaGPT, AutoGen, CrewAI) and their orchestration mechanisms.

5.3 When Does Multi-Agency Help?

Conditions under which multi-agent designs outperform monolithic agents; coordination overhead, failure modes, and diminishing returns with agent count.

6 Execution Feedback and Iterative Refinement

Closing the loop: agents execute generated code, observe runtime outcomes (test failures, compiler errors, exceptions), and revise their output. Covers both general refinement and fault-specific self-repair.

6.1 Feedback Signals and Environments

Types of feedback: test results, compiler errors, runtime exceptions, static analysis, human approval sig-

nals. Sandboxed execution environments and their design.

6.2 Iterative Refinement Mechanisms

How agents incorporate feedback: reflexion-style verbal self-critique, repair prompts, diff-based editing, and multi-turn conversation with the environment.

6.3 Fault Localization and Self-Repair

When refinement becomes debugging: stack-trace analysis, LLM-driven root-cause reasoning, edit-based patching, candidate ranking, and test-driven repair loops.

6.4 Convergence and Stopping Criteria

When to stop iterating: pass-rate thresholds, diminishing improvement detection, budget limits, and the risk of over-fitting to weak test suites.

7 Discussion

7.1 Consolidated Patterns and Trends

Recurring design patterns across all five topics (observe-think-act loops, tool orchestration, LLM-as-judge verification). Macro trends: from single-shot prompting to agentic scaffolds, from monolithic agents to composable pipelines, from flat CoT to tree-structured search.

7.2 The Evaluation Landscape

Cross-cutting view of benchmarks (HumanEval, MBPP, SWE-bench, Defects4J, CrossCodeEval), their coverage of pipeline stages, saturation concerns, and gaps in realism.

7.3 The Context Window Bottleneck

How finite context constrains every pipeline stage; retrieval-augmented generation, hierarchical summarization, and repository-level chunking as mitigations.

7.4 Open Challenges

Unsolved problems: long-horizon reasoning, specification ambiguity, cost efficiency, trustworthiness, and human-in-the-loop control design.

7.5 Implications for Software Engineering Practice

How these advances may reshape developer workflows, team structures, and the economics of software production.

8 Conclusion

Summarize key findings across the five research fronts, restate the pipeline framing as a unifying lens, and point to the most promising future directions.

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